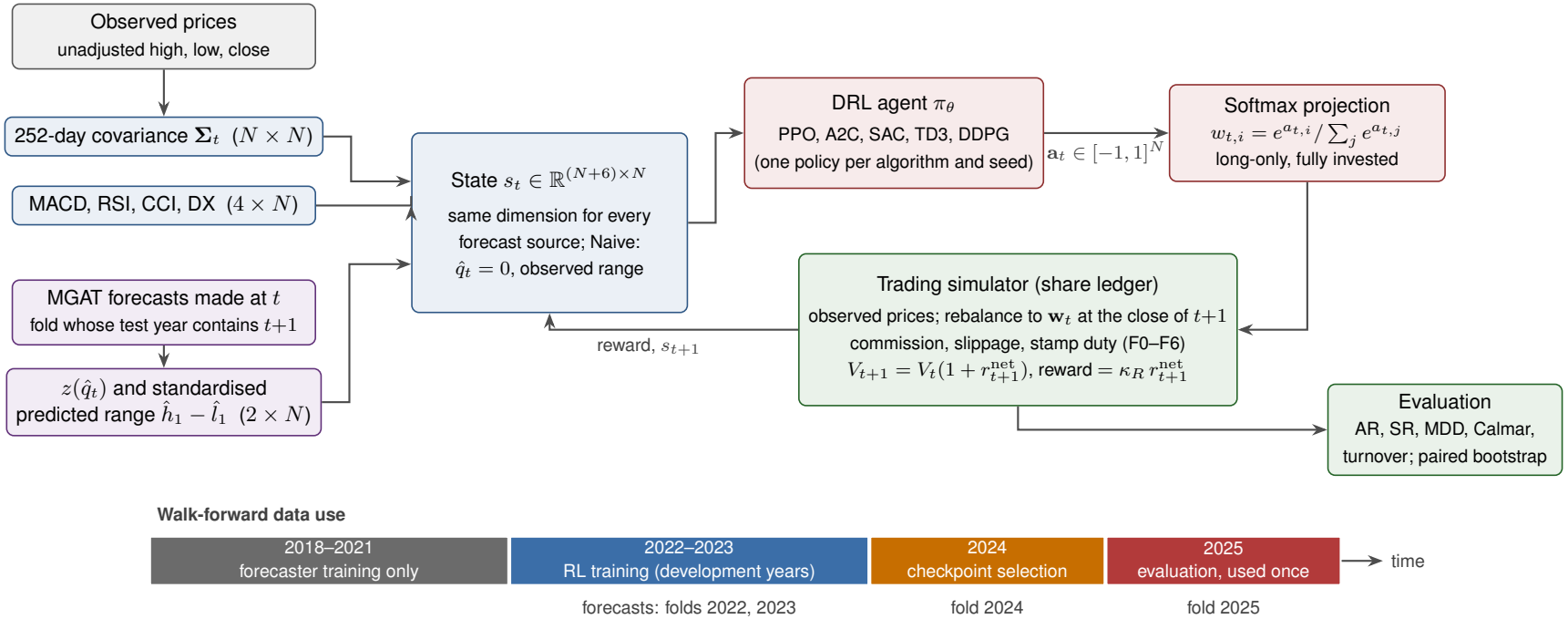

Observed prices
unadjusted high, low, close
252-day covariance $\mathbf{\Sigma}_t$ $(N \times N)$
MACD, RSI, CCI, DX $(4 \times N)$
MGAT forecasts made at t
fold whose test year contains $t{+}1$
$z(\hat{q}_t)$ and standardised predicted range $\hat{h}_1 - \hat{l}_1$ $(2 \times N)$
State $s_t \in \mathbb{R}^{(N+6)\times N}$
same dimension for every forecast source; Naive: $\hat{q}_t = 0$, observed range
DRL agent π_θ
PPO, A2C, SAC, TD3, DDPG
(one policy per algorithm and seed)
$\mathbf{a}_t \in [-1, 1]^N$
Softmax projection
$w_{t,i} = e^{a_{t,i}} / \sum_j e^{a_{t,j}}$
long-only, fully invested
Trading simulator (share ledger)
observed prices; rebalance to $\mathbf{w}_t$ at the close of $t{+}1$
commission, slippage, stamp duty (F0–F6)
$V_{t+1} = V_t(1 + r^{\text{net}}_{t+1})$, reward $= \kappa_R\, r^{\text{net}}_{t+1}$
reward, s_{t+1}
Evaluation
AR, SR, MDD, Calmar,
turnover; paired bootstrap
Walk-forward data use
2018–2021
forecaster training only
2022–2023
RL training (development years)
2024
checkpoint selection
2025
evaluation, used once
time
forecasts: folds 2022, 2023
fold 2024
fold 2025